Cost Analysis

2026-05-18

Contents

1 Assumptions

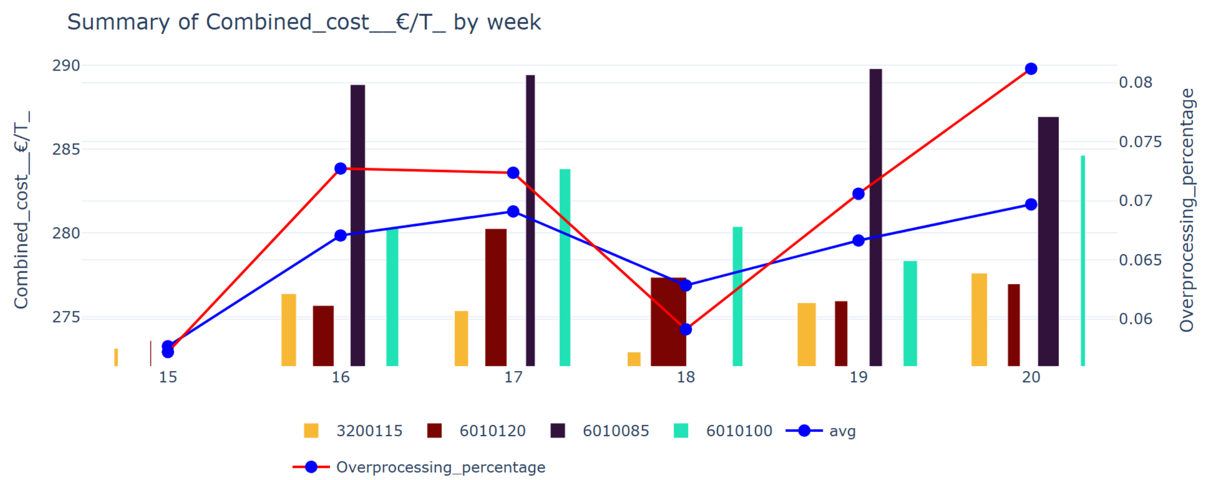
The aim of the analysis is to identify the most cost-effective production periods in months of data and recommend optimal operating windows for the corresponding key influencers that can be set as a benchmark for the weekly analysis.

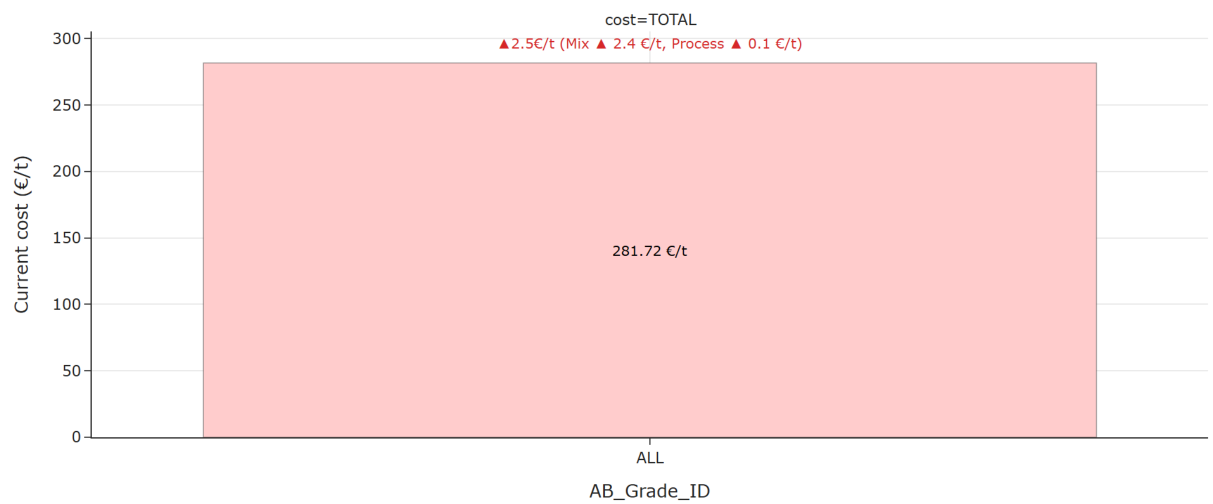
- **Period Analyzed**: April 12, 2026 – May 18, 2026
- **Cost Components Included**: Fibre, steam, electricity, starch, and chemicals.
- **Grades Included**: Only grades 6010085, 6010100, 601020 and 3200115 are included in this week's analysis.
- **Data Selection Criteria**: Only operating conditions where strength properties exceed specification thresholds were considered.
- **Naming Conventions Used**:
  - **Current**: Refers to the last week of the analyzed period.
  - **Historic**: Represents the average cost over the monthly interval previous to the "Current" period. Used as baseline.
  - **Best**: Corresponds to the cost associated with the optimal batch configuration.

2 Executive Summary

The following plot illustrates how total cost varies across different grades, enabling a comparison between the current configuration and the historical baseline.

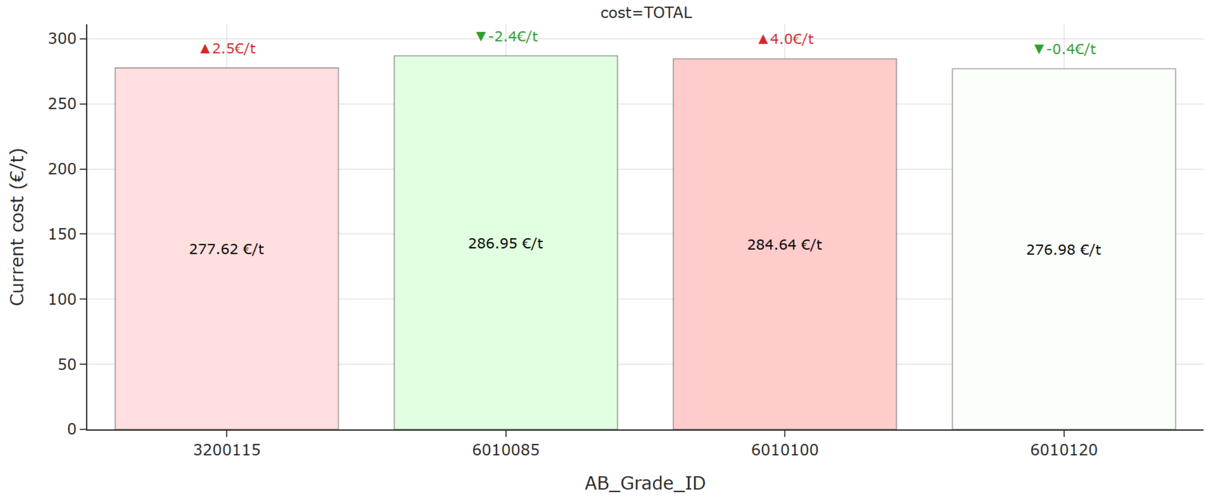
The next plot displays how total cost evolves over time, with a breakdown by grade to show individual contributions.





For all considered grades, TOTAL cost saw an increase of 2.48 €/t, resulting in 281.72 €/t.

Breaking down the overall cost change, +2.41 €/t was driven by a more expensive mix of grades, and +0.06 €/t was due to a deterioration in the process.



For grade 3200115, TOTAL cost saw an increase of 2.53 €/t, resulting in 277.62 €/t.

For grade 6010085, TOTAL cost saw a decrease of 2.39 €/t, resulting in 286.95 €/t.

For grade 6010100, TOTAL cost saw an increase of 4.05 €/t, resulting in 284.64 €/t.

For grade 6010120, TOTAL cost saw a decrease of 0.43 €/t, resulting in 276.98 €/t.

Looking across individual grades, the strongest cost increase was observed for grade 6010100 (+4.05 €/t) and the strongest cost decrease was observed for grade 6010085 (-2.39 €/t). These movements stand out and may require further investigation.

Average change by component:

- **Electricity__€/T** increased on average (+0.77 €/t).
- **Steam__€/T** increased on average (+0.52 €/t).
- **Starch__€/T** decreased on average (-0.42 €/t).
- **Fibre_cost__€/T** increased on average (+0.08 €/t).
- **Chemicals__€/T** decreased on average (-0.01 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Electricity__€/T** (+0.77 €/t on average), with the highest increase in grade 6010100 (+1.25 €/t) and the largest average decrease comes from **Starch__€/T** (-0.42 €/t on average), with the biggest drop in grade 6010085 (-1.57 €/t). These components stand out and may require further investigation.

Average change by steam:

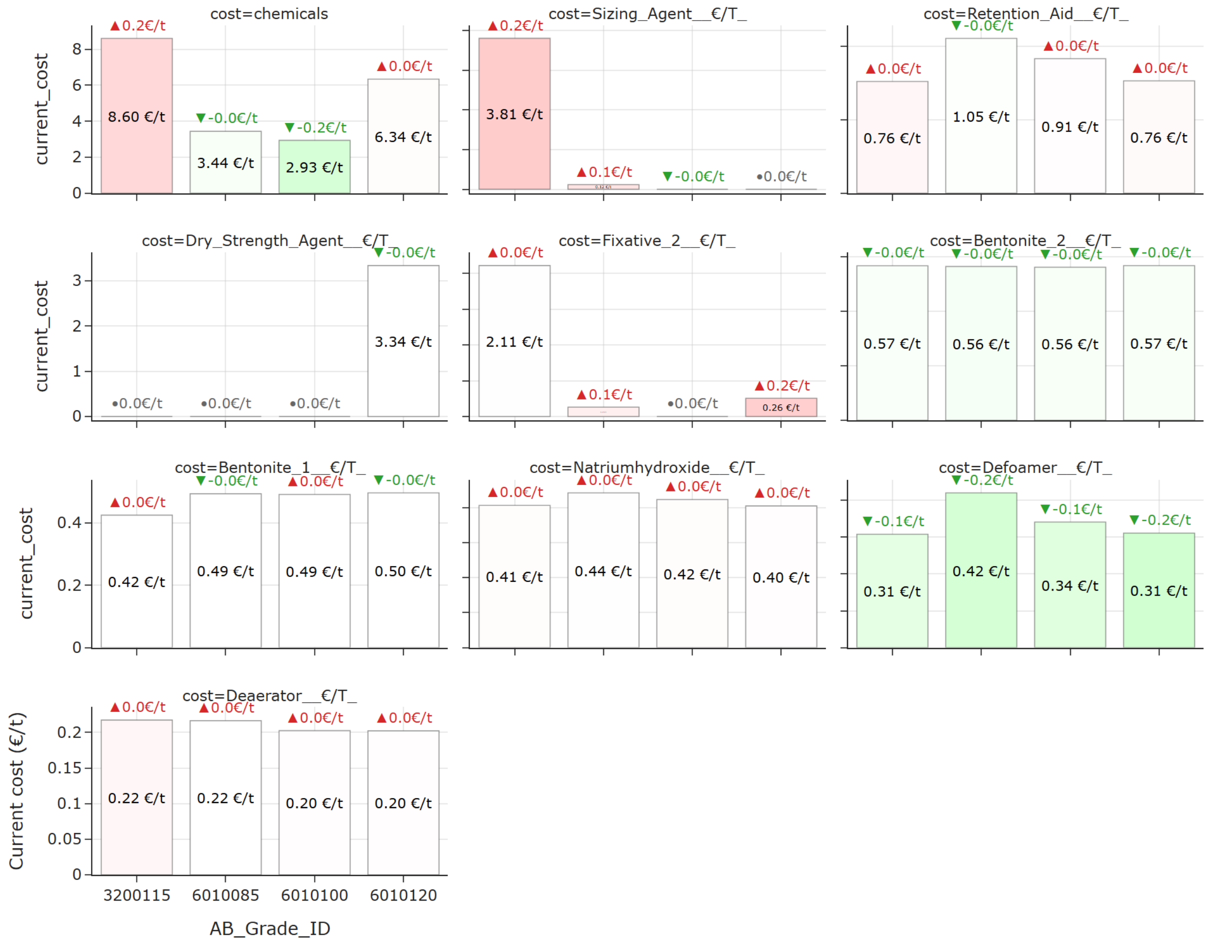
- **Steam_Whitewater** increased on average (+0.30 €/t).
- **Steam_Predryers** increased on average (+0.19 €/t).
- **Steam_Starchkitchen** increased on average (+0.04 €/t).
- **Steam_Afterdryers** decreased on average (-0.02 €/t).
- **Steam_Heatexchangers** increased on average (+0.01 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Steam_Whitewater** (+0.30 €/t on average), with the highest increase in grade 6010100 (+1.10 €/t) and the largest average decrease comes from **Steam_Afterdryers** (-0.02 €/t on average), with the biggest drop in grade 6010085 (-0.89 €/t). These components stand out and may require further investigation.

Average change by electricity:

- **Electricity_Other** increased on average (+0.43 €/t).
- **Electricity_Specific** increased on average (+0.38 €/t).
- **Electricity_Afterdryer** increased on average (+0.03 €/t).
- **Electricity_Predryer** decreased on average (-0.02 €/t).
- **Electricity_Wiresection** decreased on average (-0.02 €/t).
- **Electricity_Press_and_Steambox** decreased on average (-0.02 €/t).
- **Electricity_Freqconverters** decreased on average (-0.01 €/t).

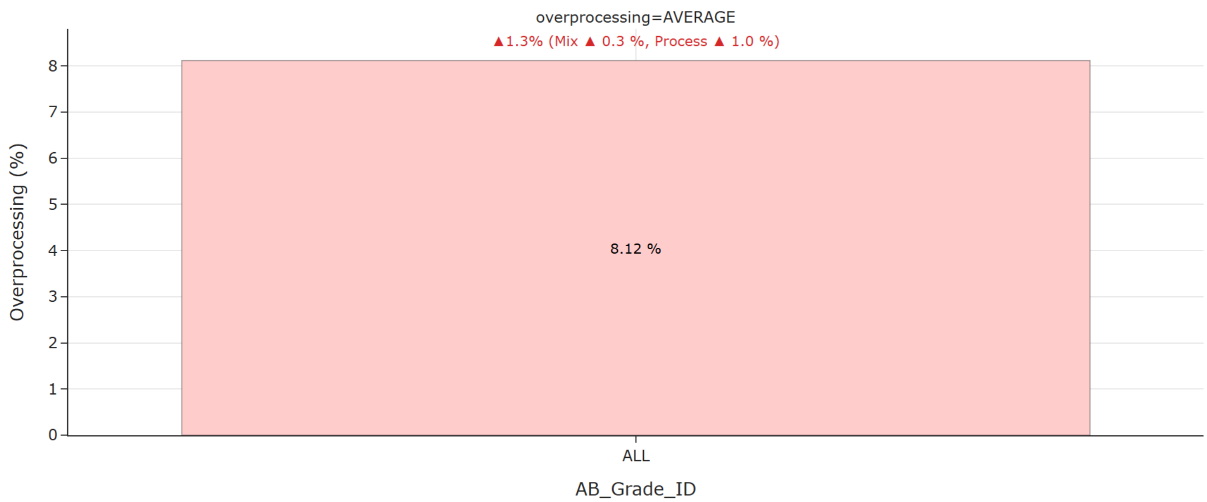
A more detailed breakdown by component shows that the largest average increase comes from **Electricity_Other** (+0.43 €/t on average), with the highest increase in grade 6010100 (+0.66 €/t) and the largest average decrease comes from **Electricity_Predryer** (-0.02 €/t on average), with the biggest drop in grade 6010085 (-0.04 €/t). These components stand out and may require further investigation.



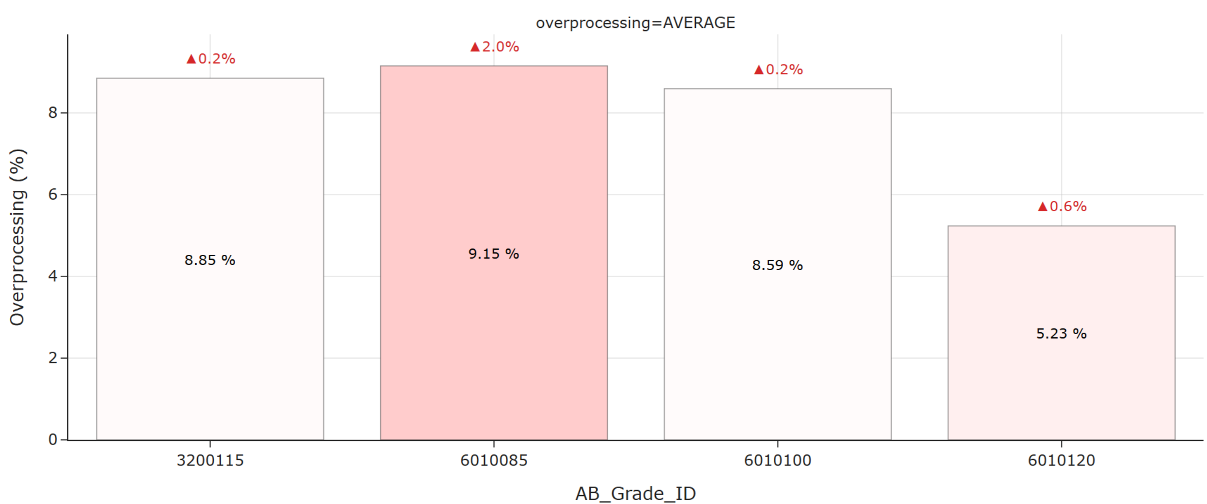
Average change by chemicals:

- **Sizing__Agent__€/T** increased on average (+0.17 €/t).
- **Defoamer__€/T** decreased on average (-0.16 €/t).
- **Fixative__2__€/T** increased on average (+0.09 €/t).
- **Bentonite__2__€/T** decreased on average (-0.03 €/t).
- **Retention__Aid__€/T** increased on average (+0.01 €/t).
- **Deaerator__€/T** increased on average (+0.01 €/t).
- **Natriumhydroxide__€/T** increased on average (+0.01 €/t).
- **Bentonite__1__€/T** decreased on average (-0.00 €/t).
- **Dry__Strength__Agent__€/T** decreased on average (-0.00 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Sizing_Agent_€/T** (+0.17 €/t on average), with the highest increase in grade 3200115 (+0.22 €/t) and the largest average decrease comes from **Defoamer_€/T** (-0.16 €/t on average), with the biggest drop in grade 6010120 (-0.20 €/t). These components stand out and may require further investigation.



For all considered grades, TOTAL overprocessing saw an increase of 1.27 %, resulting in 8.12 %.



For grade 3200115, TOTAL overprocessing saw an increase of 0.20 %, resulting in 8.85 %.

For grade 6010085, TOTAL overprocessing saw an increase of 1.96 %, resulting in 9.15 %.

For grade 6010100, TOTAL overprocessing saw an increase of 0.15 %, resulting in 8.59 %.

For grade 6010120, TOTAL overprocessing saw an increase of 0.61 %, resulting in 5.23 %.

Looking across individual grades, the strongest overprocessing increase was observed for grade

6010085 (+1.96 %). These movements stand out and may require further investigation.

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## AttributeError: 'DataFrame' object has no attribute 'cost'. Did you mean: 'count'?
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For grade 3200115, TOTAL overprocessing saw an increase of 0.20 %, resulting in 8.85 %.

For grade 6010085, TOTAL overprocessing saw an increase of 1.96 %, resulting in 9.15 %.

For grade 6010100, TOTAL overprocessing saw an increase of 0.15 %, resulting in 8.59 %.

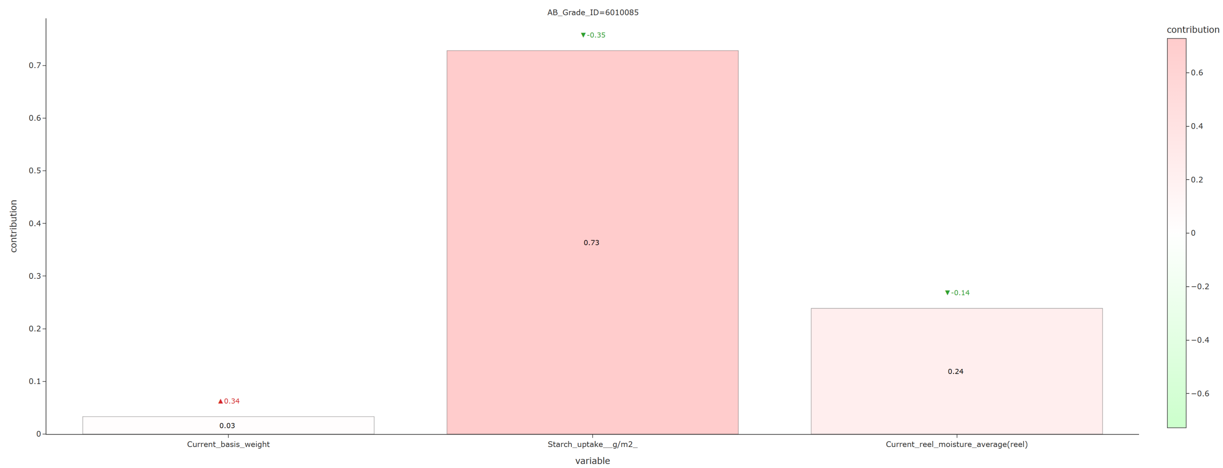
For grade 6010120, TOTAL overprocessing saw an increase of 0.61 %, resulting in 5.23 %.

Looking across individual grades, the strongest overprocessing increase was observed for grade 6010085 (+1.96 %). These movements stand out and may require further investigation.

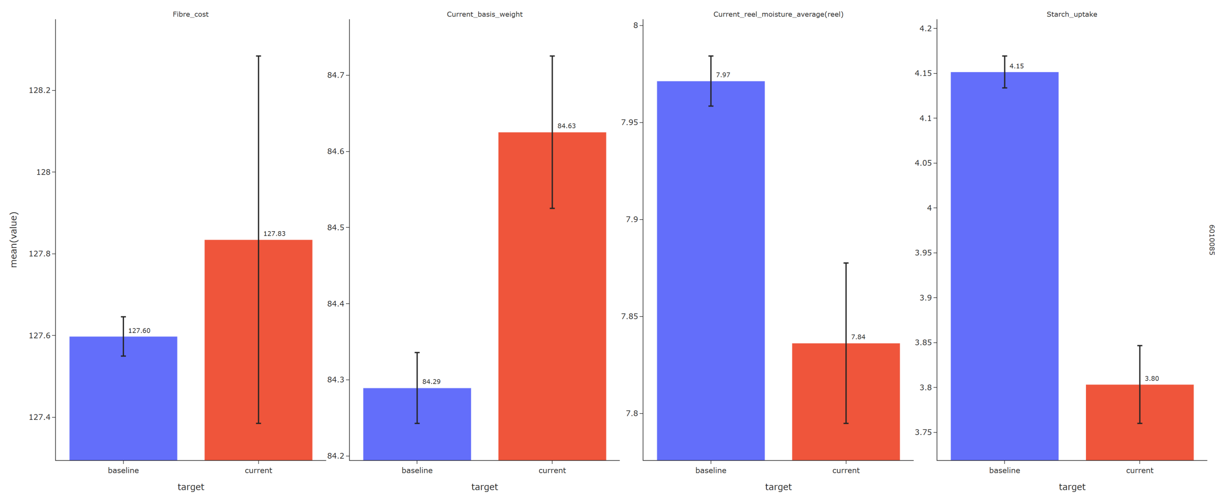
3 Breakdown by grades

3.1 6010085

3.1.1 Fiber cost

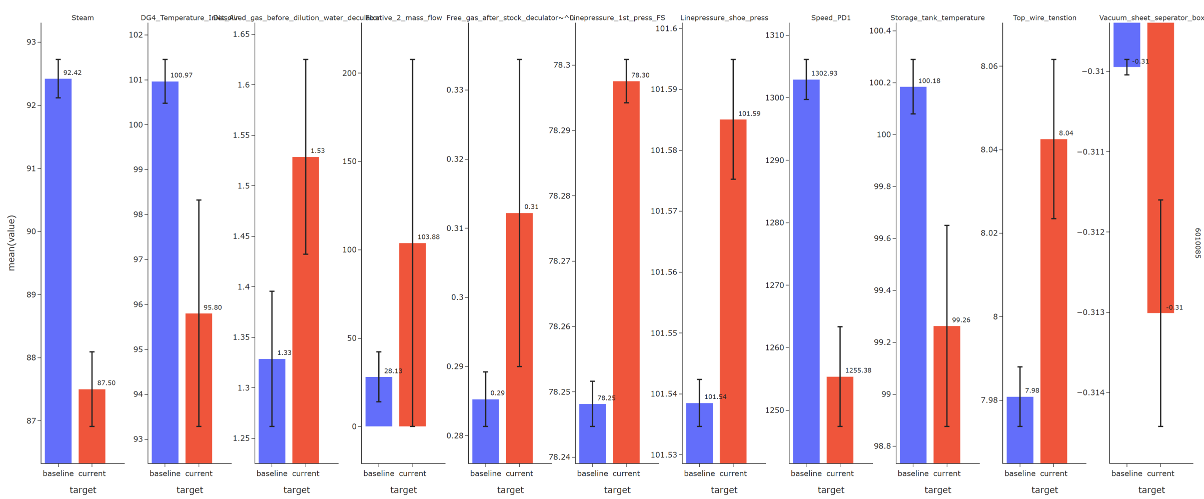


For grade 6010085, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-increasing drivers: - Starch uptake g/m2 decreased (-0.35)

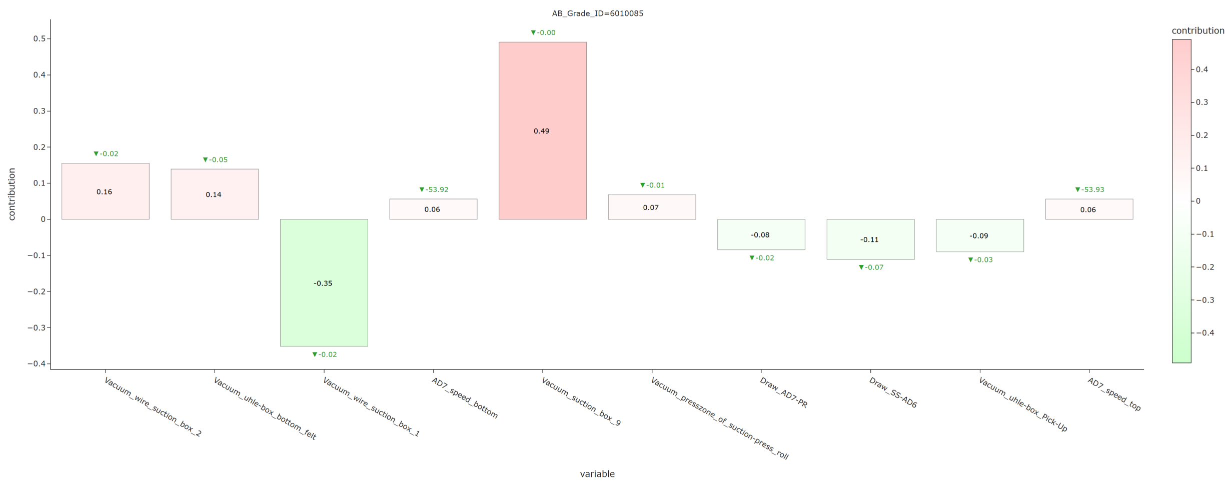


3.1.2 Steam cost

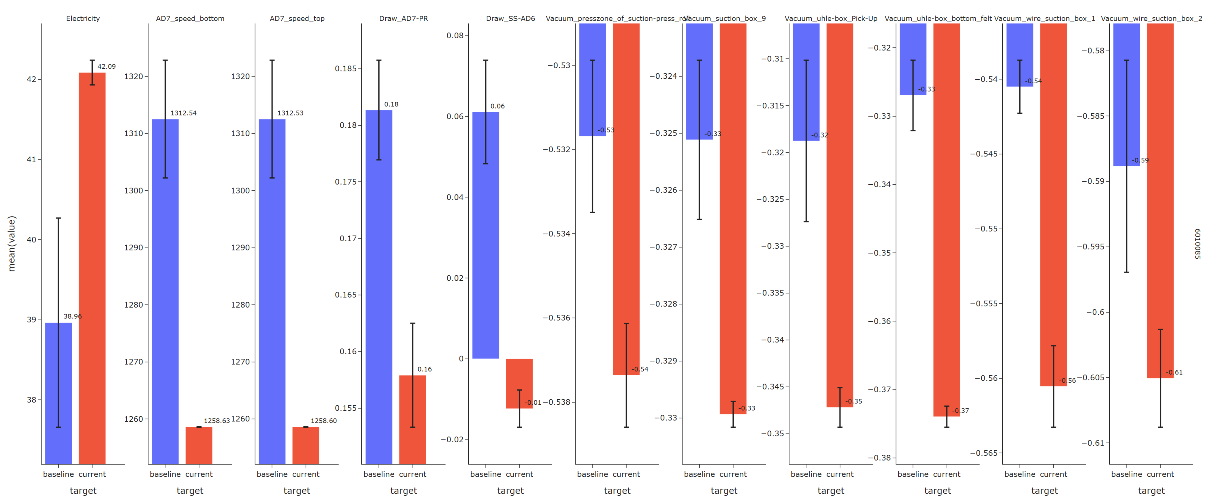
For grade 6010085, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Top wire tenstion increased (+0.06) - Linepressure 1st press FS bar increased (+0.05) - Dissolved gas before dilution water deculator increased (+0.20) - Fixative 2 mass flow g/T increased (+75.75) - Linepressure shoe press bar increased (+0.05) Cost-increasing drivers: - Storage tank temperature decreased (-0.92) - Vacuum sheet seperator box decreased (-0.00)



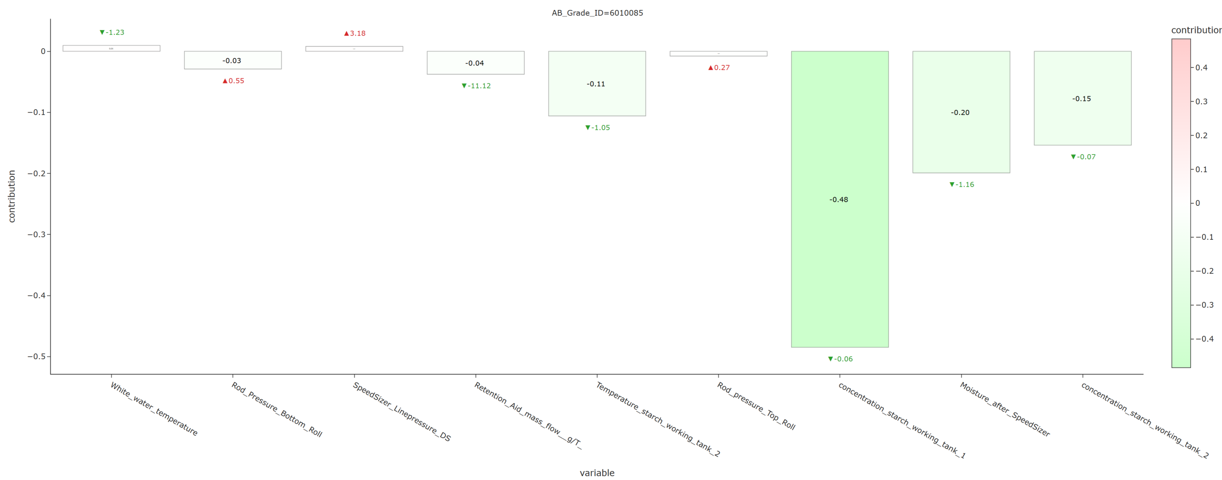
3.1.3 Electricity cost



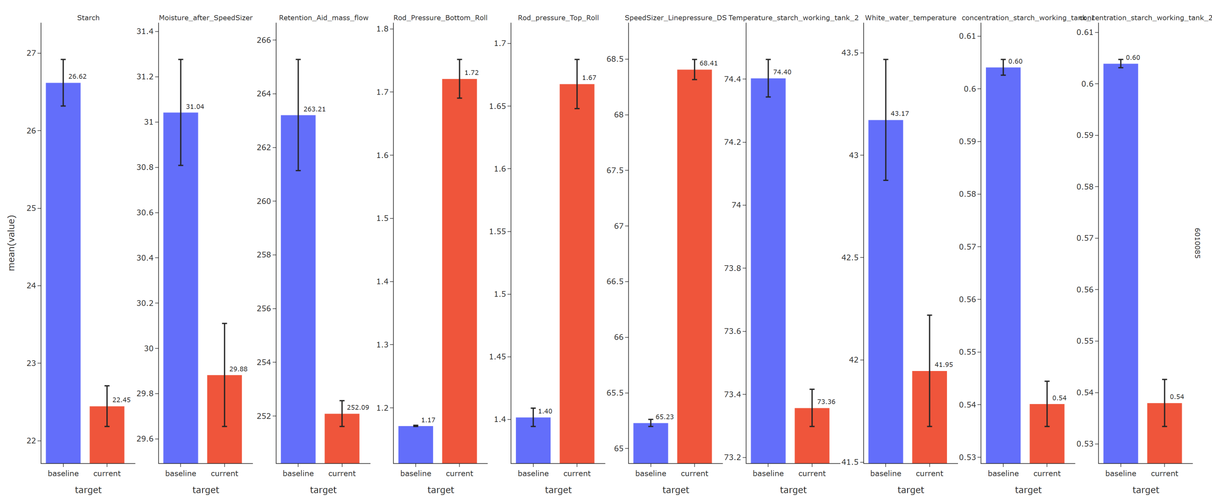
For grade 6010085, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Vacuum wire suction box 1 decreased (-0.02) - Draw SS-AD6 decreased (-0.07) - Vacuum uhle-box Pick-Up decreased (-0.03) Cost-increasing drivers: - Vacuum suction box 9 decreased (-0.00) - Vacuum wire suction box 2 decreased (-0.02) - Vacuum uhle-box bottom felt decreased (-0.05)



3.1.4 Starch cost

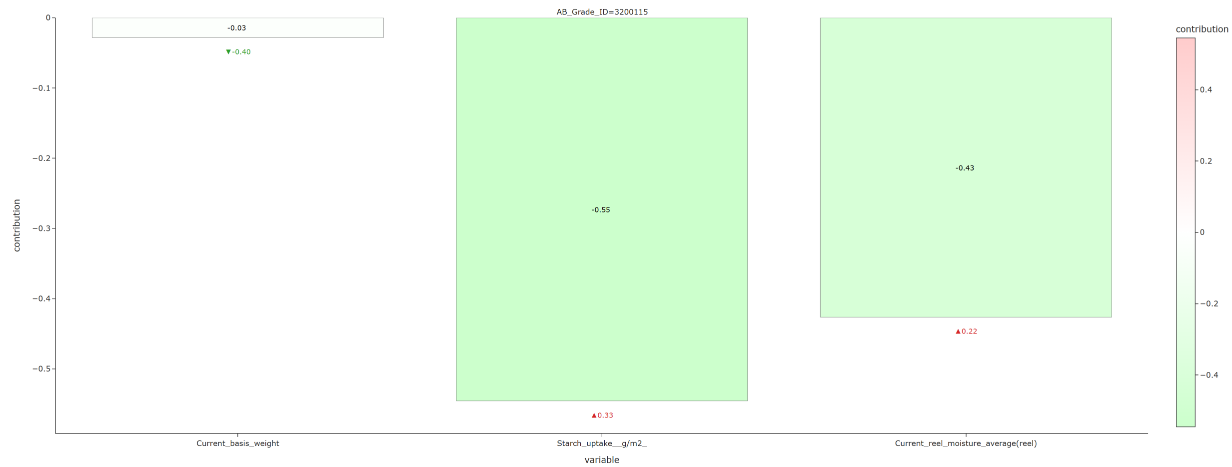


For grade 6010085, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Concentration starch working tank 1 decreased (-0.06) - Moisture after SpeedSizer decreased (-1.16)

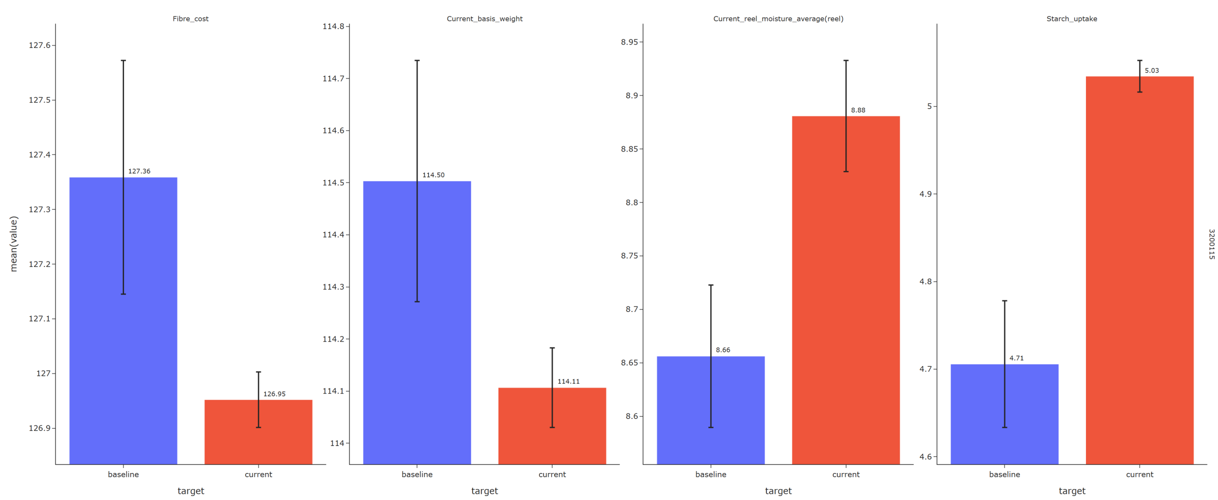


3.2 3200115

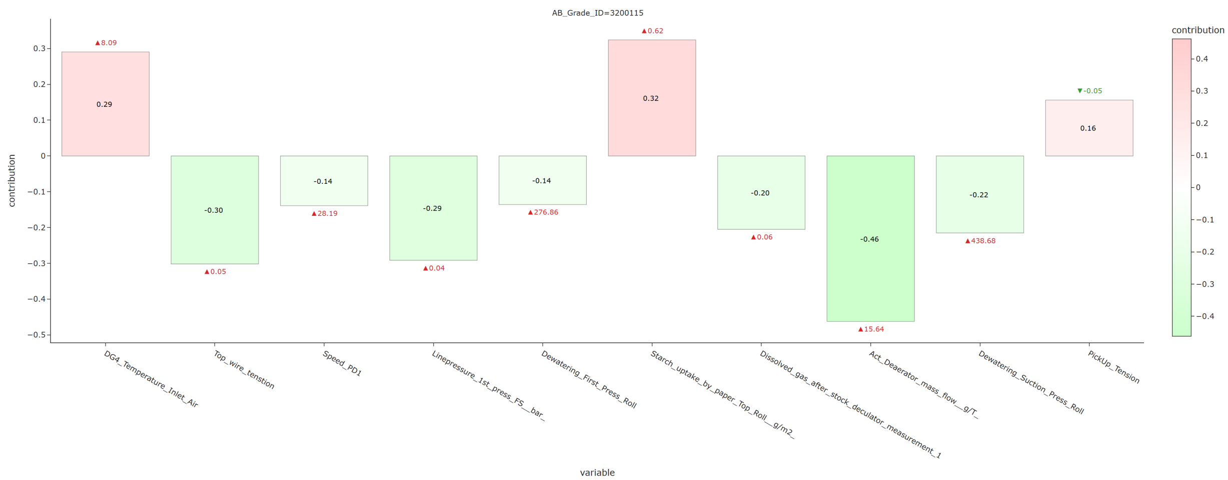
3.2.1 Fiber cost



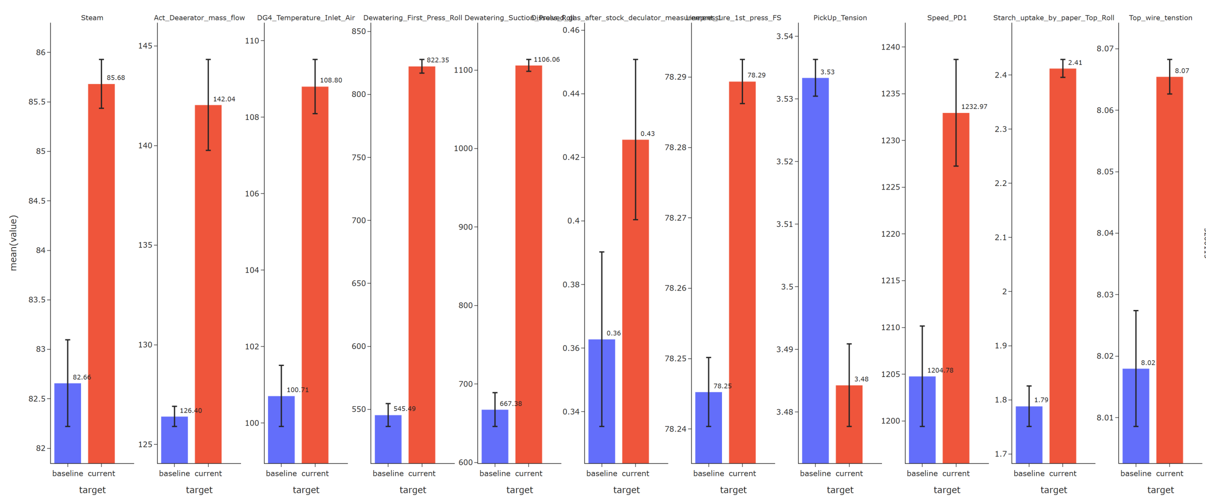
For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Starch uptake g/m2 increased (+0.33)



3.2.2 Steam cost

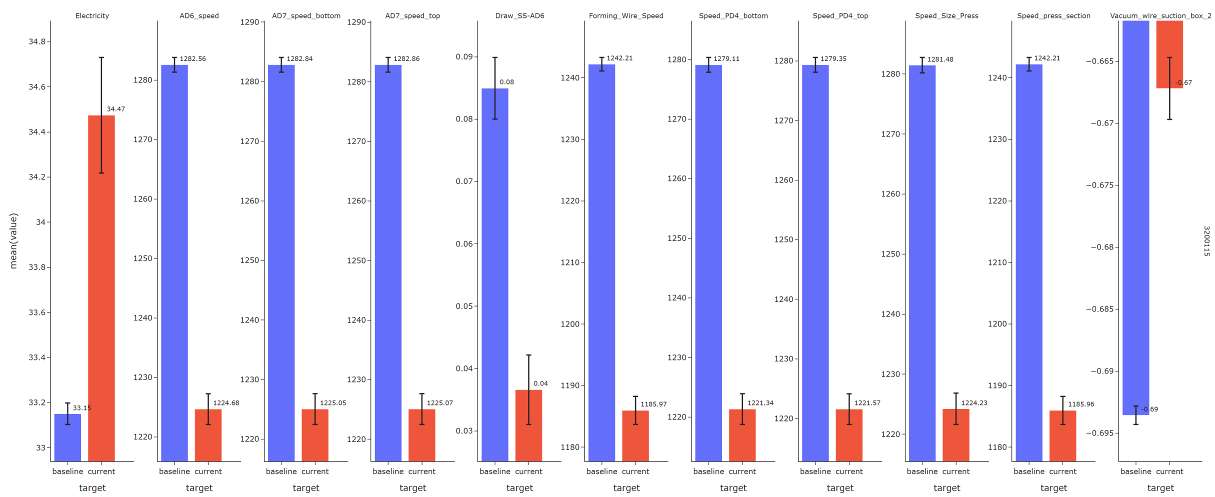


For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Act Deaerator mass flow g/T increased (+15.64) - Top wire tenstion increased (+0.05) - Linepressure 1st press FS bar increased (+0.04) Cost-increasing drivers: - Starch uptake by paper Top Roll g/m2 increased (+0.62) - DG4 Temperature Inlet Air increased (+8.09)

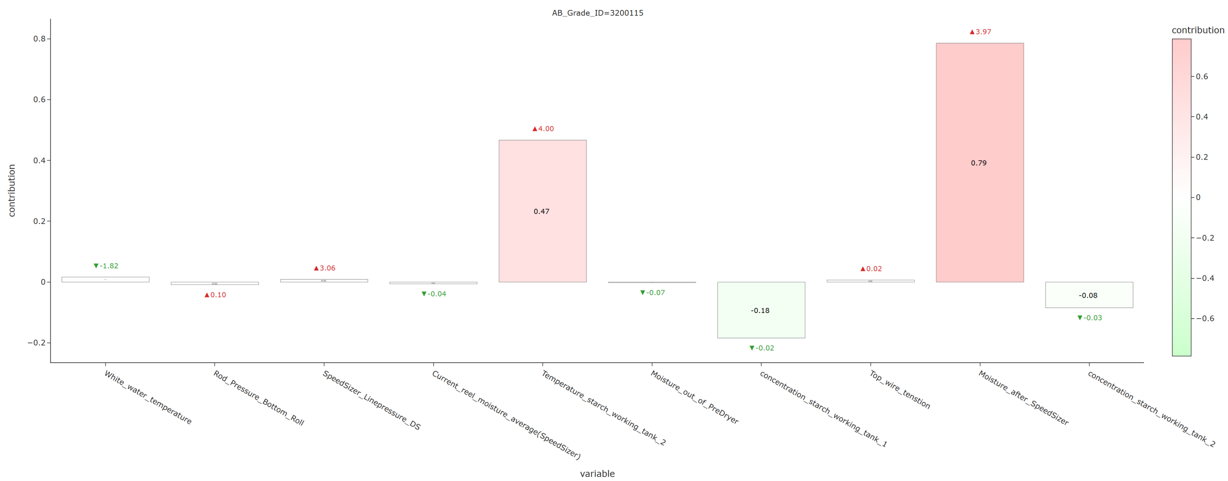


3.2.3 Electricity cost

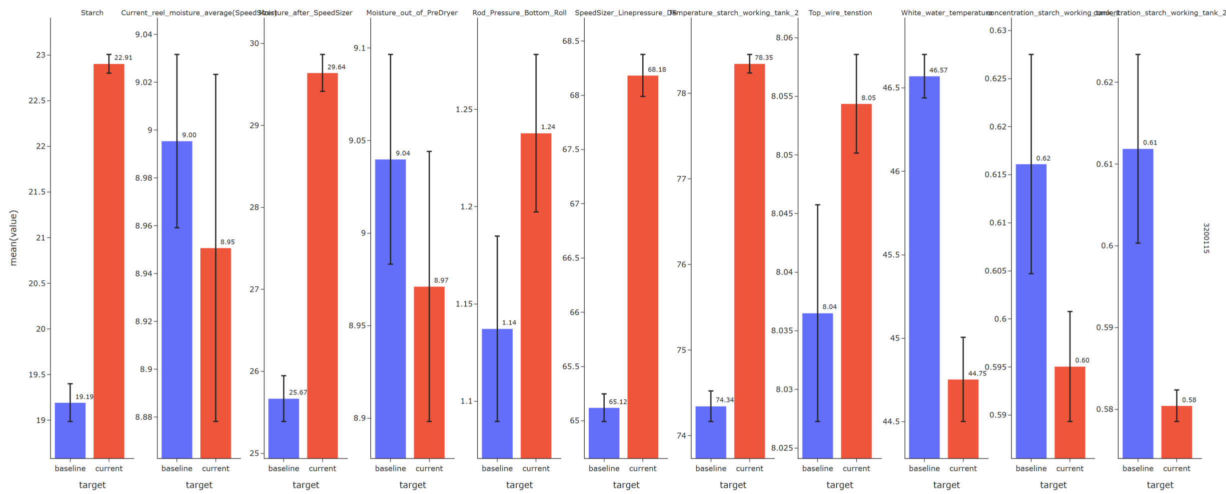
For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Vacuum wire suction box 2 increased (+0.03) - Draw SS-AD6 decreased (-0.05) Cost-increasing drivers: - Speed press section decreased (-56.25) - Forming Wire Speed decreased (-56.24) - Speed PD4 top decreased (-57.78)



3.2.4 Starch cost



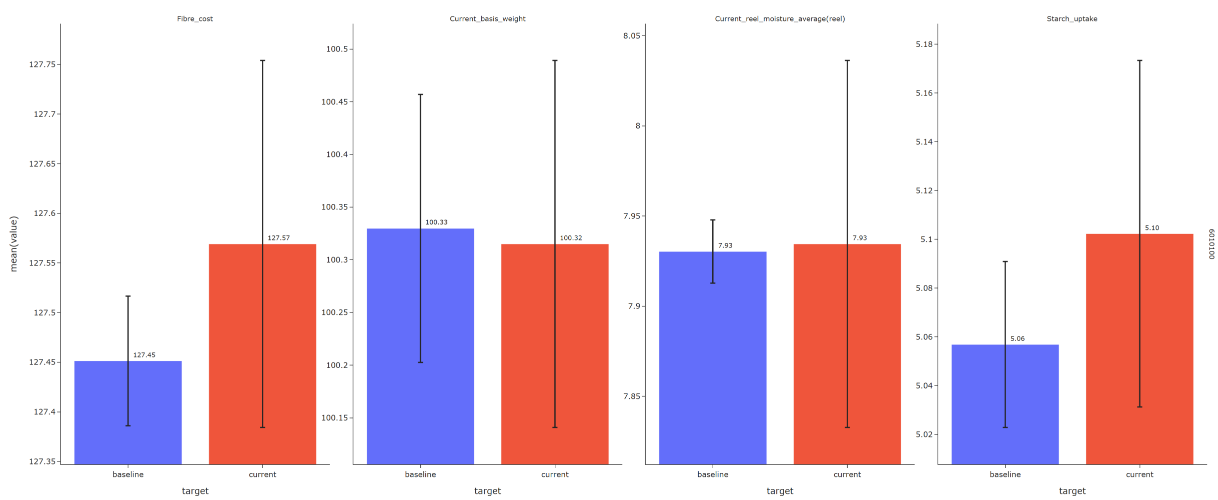
For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-increasing drivers: - Moisture after SpeedSizer increased (+3.97) - Temperature starch working tank 2 increased (+4.00)



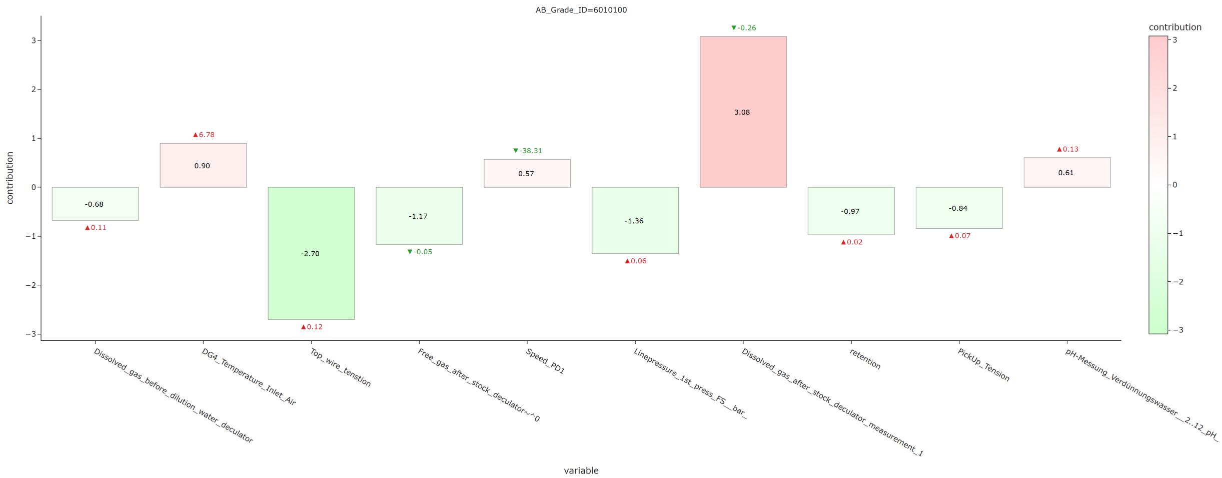
3.3 6010100

3.3.1 Fiber cost

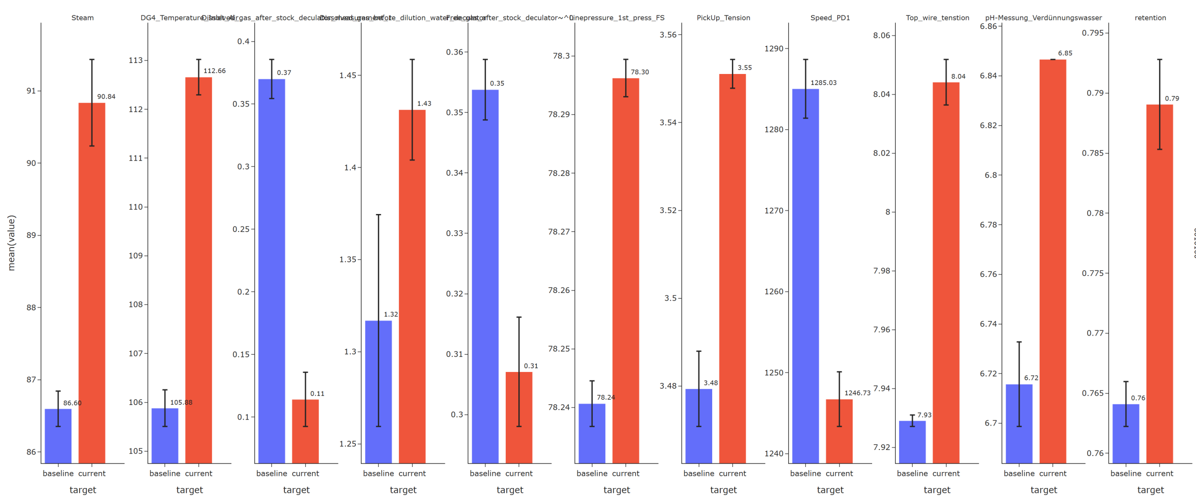
For grade 6010100, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Starch uptake g/m2 increased (+0.05)



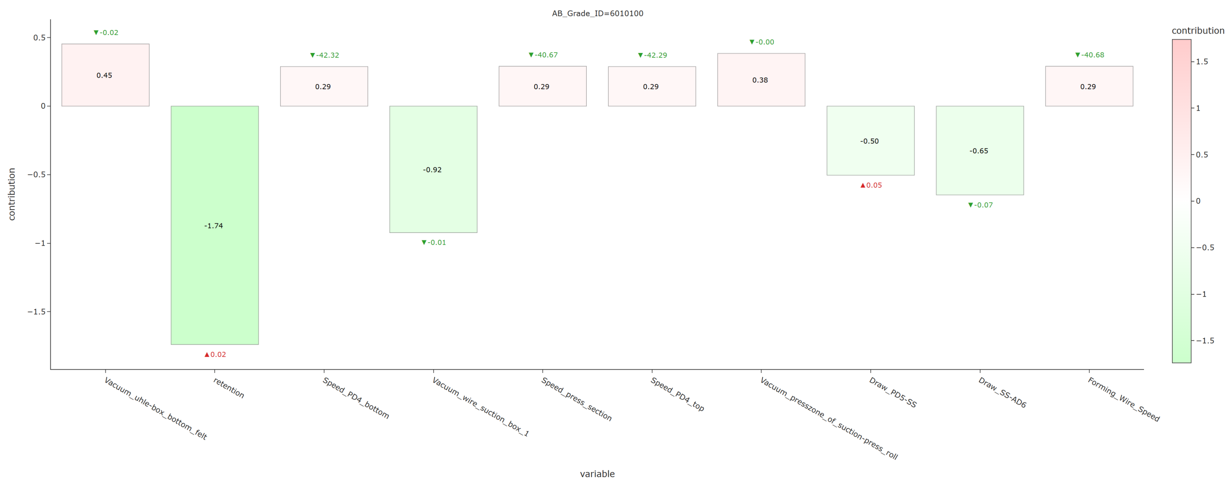
3.3.2 Steam cost



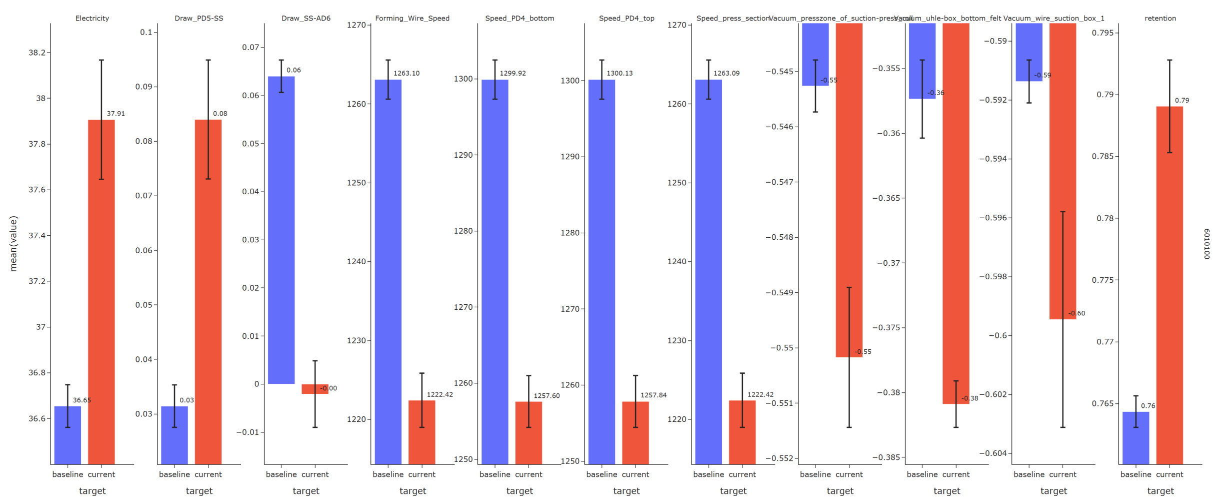
For grade 6010100, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Top wire tenstion increased (+0.12) - Linepressure 1st press FS bar increased (+0.06) - Free gas after stock deculator~^0 decreased (-0.05) - Retention increased (+0.02) Cost-increasing drivers: - Dissolved gas after stock deculator measurement 1 decreased (-0.26) - DG4 Temperature Inlet Air increased (+6.78)



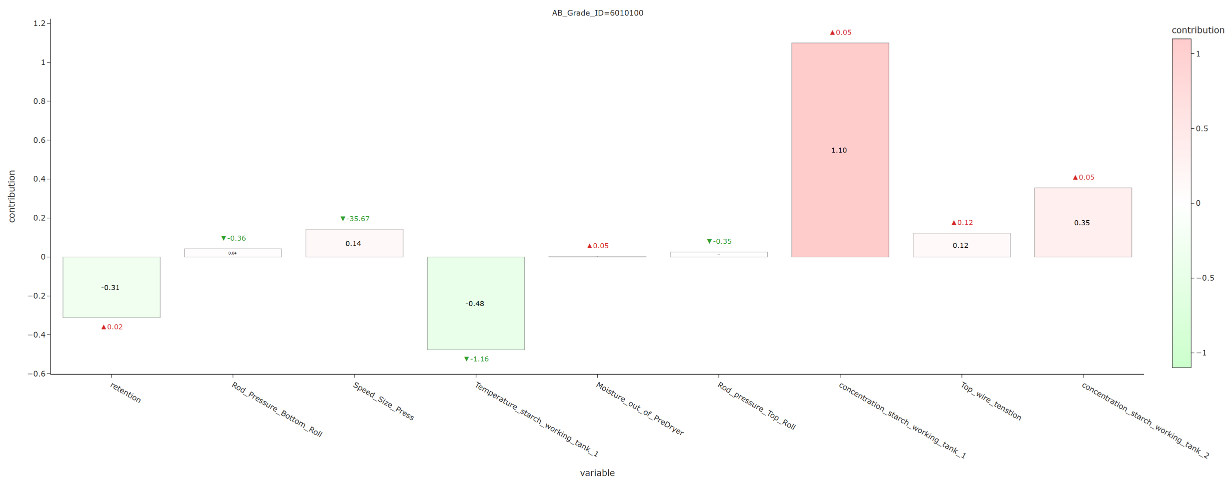
3.3.3 Electricity cost



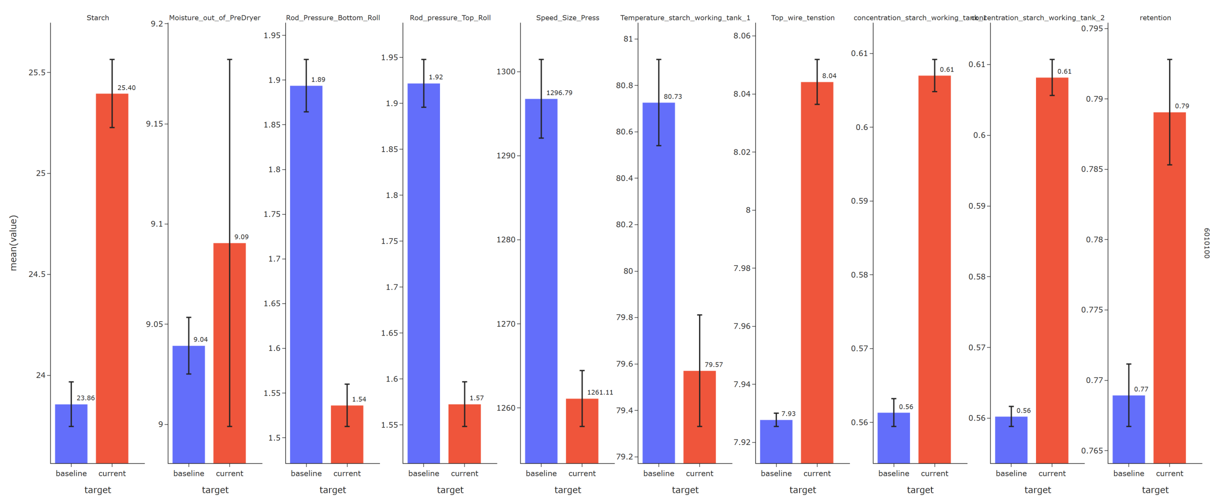
For grade 6010100, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Retention increased (+0.02) - Vacuum wire suction box 1 decreased (-0.01) - Draw SS-AD6 decreased (-0.07) - Draw PD5-SS increased (+0.05) Cost-increasing drivers: - Vacuum uhle-box bottom felt decreased (-0.02) - Vacuum presszone of suction-press roll decreased (-0.00)



3.3.4 Starch cost



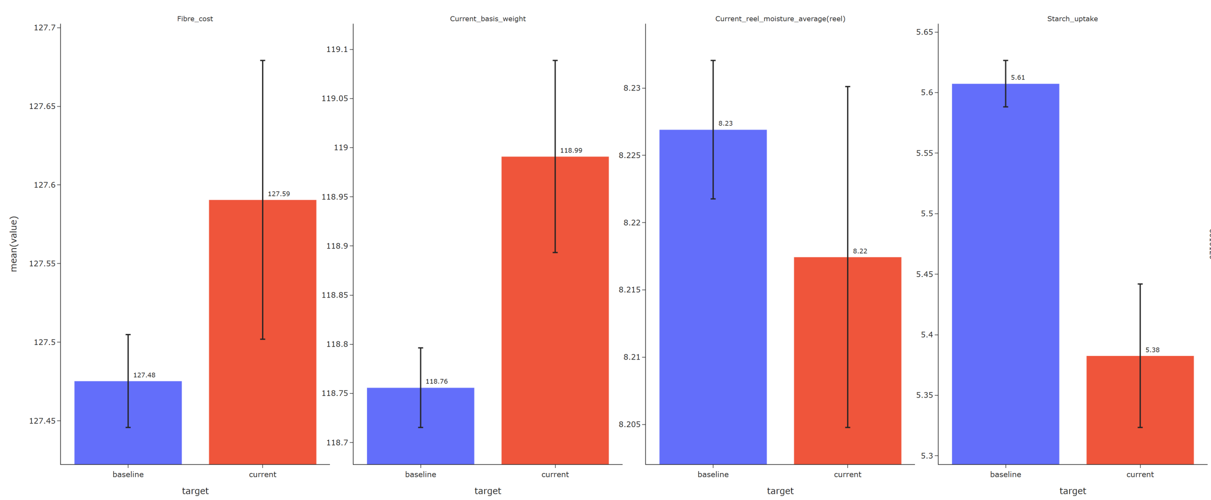
For grade 6010100, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Temperature starch working tank 1 decreased (-1.16) Cost-increasing drivers: - Concentration starch working tank 1 increased (+0.05)



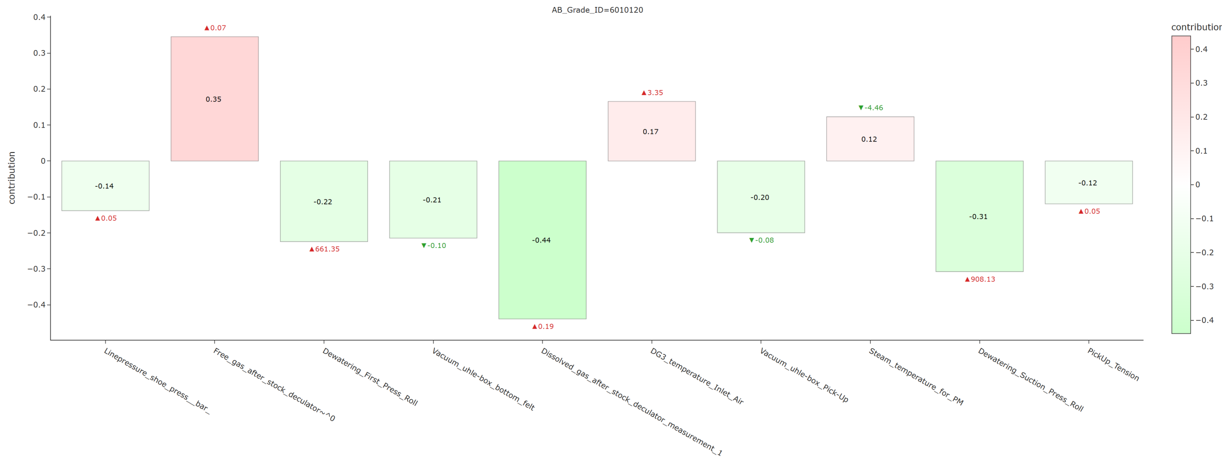
3.4 6010120

3.4.1 Fiber cost

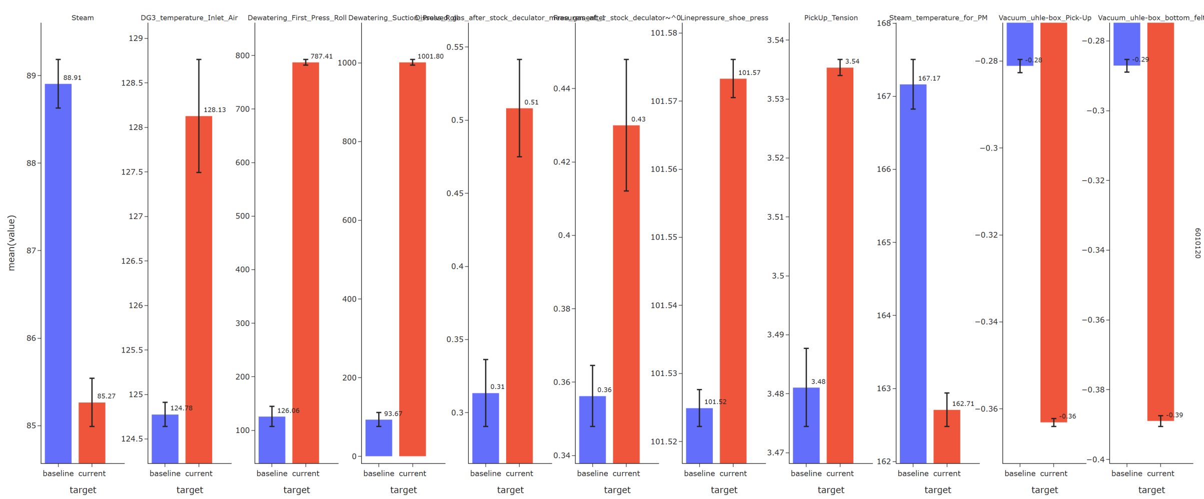
For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-increasing drivers: - Starch uptake g/m2 decreased (-0.22)



3.4.2 Steam cost

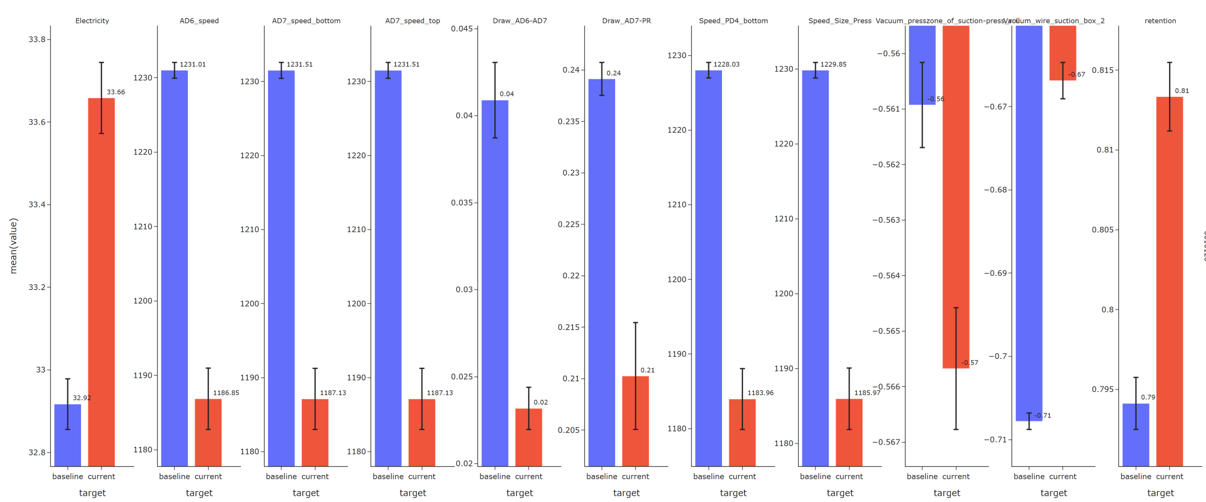


For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Dissolved gas after stock deculator measurement 1 increased (+0.19) - Dewatering Suction Press Roll increased (+908.13) - Dewatering First Press Roll increased (+661.35) - Vacuum uhle-box bottom felt decreased (-0.10) - Vacuum uhle-box Pick-Up decreased (-0.08) Cost-increasing drivers: - Free gas after stock deculator~^0 increased (+0.07)

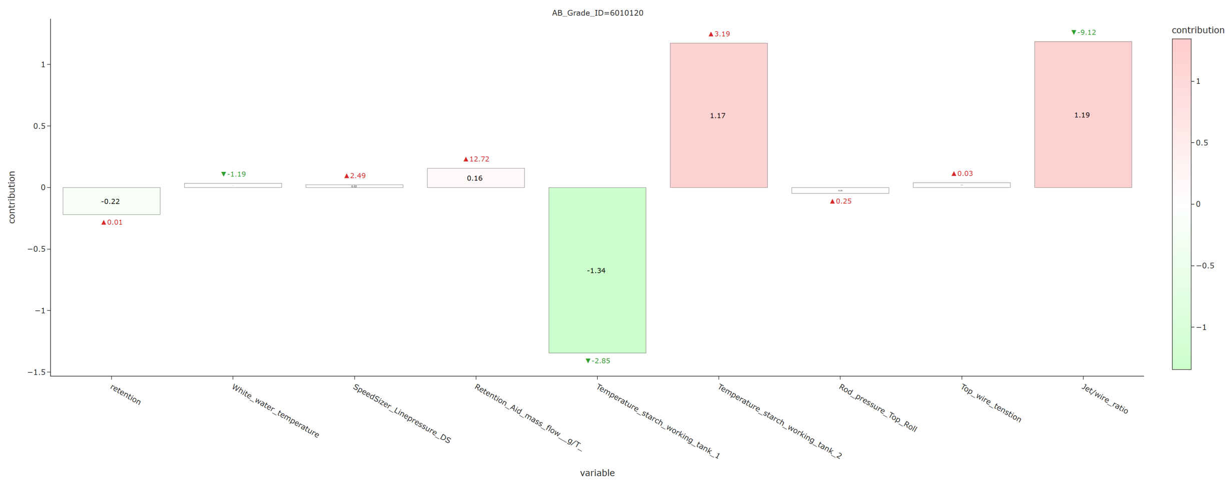


3.4.3 Electricity cost

For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Vacuum wire suction box 2 increased (+0.04) - Retention increased (+0.02) - Draw AD7-PR decreased (-0.03) Cost-increasing drivers: - Vacuum presszone of suction-press roll decreased (-0.00) - Draw AD6-AD7 decreased (-0.02)



3.4.4 Starch cost



For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Temperature starch working tank 1 decreased (-2.85) Cost-increasing drivers: - Jet/wire ratio decreased (-9.12)